数值算法 Homework 08

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Problem 1

Show that a quadratic Bézier curve is part of a parabola.

Solution:
不失一般性，给定三个互不相同的控制点 $P_0 = [x_0, y_0]^{\mathrm{T}},\ P_1 = [x_1, y_1]^{\mathrm{T}},\ P_2 = [x_2, y_2]^{\mathrm{T}}$
二次 Bézier 曲线为:

$$\begin{aligned}
\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} &= B(t) \\
&= (1-t)^2 P_0 + 2t(1-t) P_1 + t^2 P_2 \\
&= \begin{bmatrix} (1-t)^2 x_0 + 2t(1-t) x_1 + t^2 x_2 \\ (1-t)^2 y_0 + 2t(1-t) y_1 + t^2 y_2 \end{bmatrix} \\
&= \begin{bmatrix} (x_0 - 2x_1 + x_2)t^2 + 2(x_1 - x_0)t + x_0 \\ (y_0 - 2y_1 + y_2)t^2 + 2(y_1 - y_0)t + y_0 \end{bmatrix}
\end{aligned}$$

记 $\begin{cases} A = x_0 - 2x_1 + x_2,\ B = 2(x_1 - x_0),\ C = x_0 \\ D = y_0 - 2y_1 + y_2,\ \ E = 2(y_1 - y_0),\ F = y_0 \end{cases}$ 则我们有:

$$\begin{cases} x = At^2 + Bt + C \\ y = Dt^2 + Et + D \end{cases}$$

根据 Cramer 法则我们有:

$$\begin{aligned}
t^2 &= \frac{(y-F)E - (x-C)B}{AE - BD} \\
t &= \frac{A(y-F) - D(x-C)}{AE - BD}
\end{aligned}$$

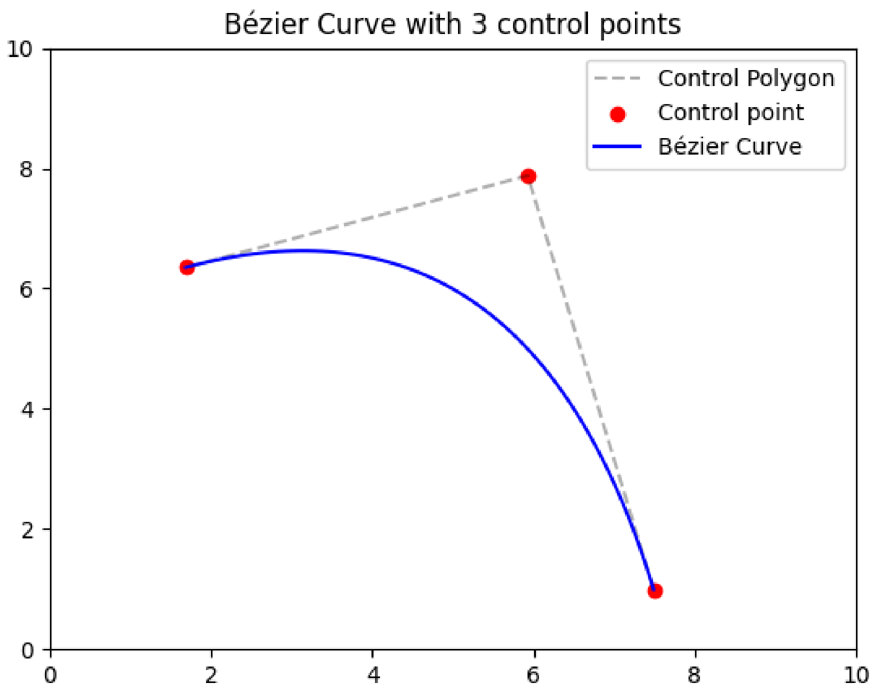
消去 t 便得到隐函数:

$$\begin{gathered}
[A(y-F) - D(x-C)]^2 = (AE - BD)[E(y-F) - B(x-C)] \\
\Updownarrow \\
D^2x^2 - 2ADxy + A^2y^2 + (2ADF - B^2D - 2CD^2)x + (-2A^2F + 2ACD - AE^2 + BDE)y \\
+(AF^2 - ABCE - 2ACDF + AE^2F + B^2CD - BDEF + C^2D^2) = 0
\end{gathered}$$

考虑其判别式:

$$\begin{aligned}
(-2AD)^2 - 4 \cdot D^2 \cdot A^2 &= 4A^2D^2 - 4A^2D^2 \\
&= 0
\end{aligned}$$

因此该二次曲线为抛物线 (parabola) 的一部分 (假设不考虑退化情况)



Problem 2

Write a program to plot the Bézier curve with arbitrary set of control points.
Visualize a few examples.

Solution:

(Casteljau 作图定理, 数值逼近, 定理 3.6)
设 $P_i = [x_i, y_i]^{\mathrm{T}}\ (i = 0, 1, \dots, n)$ 是平面上由左到右排列的 $n+1$ 个**控制点**.
定义 $P_{i,0} := P_i\ (i = 0, 1, \dots, n)$
对于任意 $t \in (0, 1)$，依次构造:

$$
\begin{aligned}
P_{i,1} &:= (1-t)P_{i,0} + tP_{i+1,0} && i = 0, 1, \dots, n-1 \\
P_{i,2} &:= (1-t)P_{i,1} + tP_{i+1,1} && i = 0, 1, \dots, n-2 \\
&\cdots \\
P_{i,k} &:= (1-t)P_{i,k-1} + tP_{i+1,k-1} && i = 0, 1, \dots, n-k \\
&\cdots \\
P_{i,n-1} &:= (1-t)P_{i,n-2} + tP_{i+1,n-2} && i = 0, 1 \\
P_{0,n} &:= (1-t)P_{0,n-1} + tP_{1,n-1}
\end{aligned}
$$

可以证明 Bézier 曲线上的点 $B(t) = P_{0,n}$ 且 $B'(t) = n(P_{1,n-1} - P_{0,n-1})$.
换言之，对于任意 $t \in (0, 1)$，$B(t)$ 的位置可以这样得到:

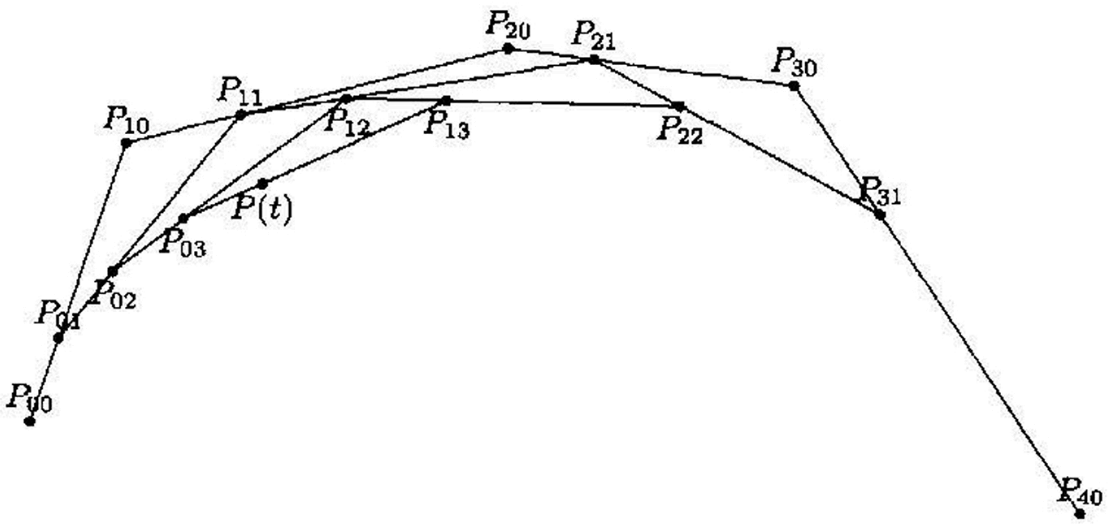


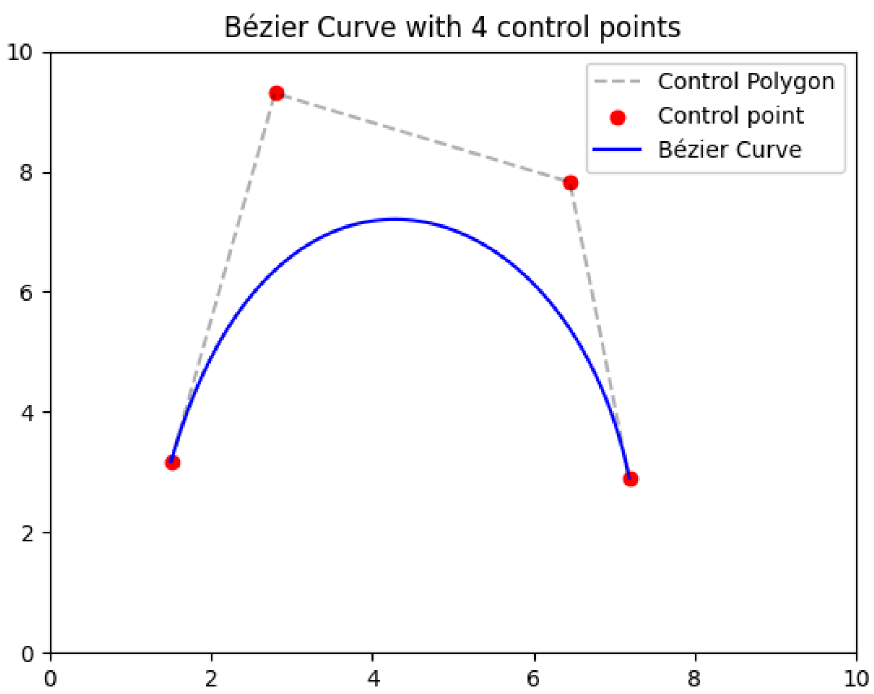
图 3.2

Python 实现:

```python
import numpy as np
import matplotlib.pyplot as plt

def bezier_point(points, t):
    current = points.copy()
    n = len(current)
    for k in range(1, n):
        for i in range(n - k):
            current[i] = (1 - t) * current[i] + t * current[i + 1]
    return current[0]

# 用户输入控制点数量
n = int(input("Number of control points: "))
np.random.seed(51)
points = np.random.rand(n, 2) * 10  # 在0-10范围内生成随机点

fig, ax = plt.subplots()
ax.set_title(f"Bézier Curve with {n} control points")
ax.set_xlim(0, 10)
ax.set_ylim(0, 10)

# 添加控制点连线（虚线灰色）
control_line, = ax.plot(points[:, 0], points[:, 1], 'k--', alpha=0.3, label='Control Polygon')
scatter = ax.scatter(points[:, 0], points[:, 1], c='red', picker=5, label='Control point')
bezier_line, = ax.plot([], [], 'b-', label='Bézier Curve')
ax.legend()

selected_point = None

def on_pick(event):
    global selected_point
    if event.artist != scatter:
        return
    selected_point = event.ind[0]
    fig.canvas.mpl_connect('motion_notify_event', on_motion)
    fig.canvas.mpl_connect('button_release_event', on_release)
```

```
def on_motion(event):
    if selected_point is None or event.inaxes != ax:
        return
    points[selected_point] = [event.xdata, event.ydata]
    scatter.set_offsets(points)
    update_curve()
    fig.canvas.draw_idle()

def on_release(event):
    global selected_point
    selected_point = None

def update_curve():
    # 更新控制点连线
    control_line.set_data(points[:, 0], points[:, 1])

    # 更新贝塞尔曲线
    if len(points) >= 2:
        t_values = np.linspace(0, 1, 100)
        curve = np.array([bezier_point(points, t) for t in t_values])
        bezier_line.set_data(curve[:, 0], curve[:, 1])
    else:
        bezier_line.set_data([], [])

fig.canvas.mpl_connect('pick_event', on_pick)
update_curve()
plt.show()
```

运行结果: (以控制点个数 $n = 4$ 为例)



Problem 3

Part (a)

Determine the degree of exactness of the quadrature rule:

$$\int_a^b f(x)\mathrm{d}x \approx \frac{b-a}{2}\left(f\left(\frac{2a+b}{3}\right)+f\left(\frac{a+2b}{3}\right)\right)$$

- **Lemma: 中点公式 $I_0(f) := (b-a)f\left(\frac{b-a}{2}\right)$ 的截断误差分析:**
  设 f 在 $[a,b]$ 上二阶连续可导.
  在中点 $c=\frac{a+b}{2}$ 处对 f 进行二阶 Taylor 展开:

  $$f(x)=f(c)+f'(c)(x-c)+\frac{f''(\eta_x)}{2}(x-c)^2 \text{ for some } \eta_x \in (a,b)$$

  左右同时在 $[a,b]$ 上积分，可得:

  $$\begin{aligned}\int_a^b f(x)\mathrm{d}x &= \int_a^b \left\{f(c)+f'(c)(x-c)+\frac{f''(\eta_x)}{2}(x-c)^2\right\}\mathrm{d}x \\ &= f(c)(b-a)+f'(c)\cdot\frac{1}{2}(x-c)^2\Big|_a^b+\frac{1}{2}\int_a^b f''(\eta_x)(x-c)^2\mathrm{d}x \quad \text{(use Mean Value Theorem for Integrals)} \\ &= I_0(f)+f'(c)\cdot 0+\frac{1}{2}f''(\xi)\int_a^b (x-c)^2\mathrm{d}x \text{ for some } \xi\in(a,b) \\ &= I_0(f)+0+\frac{1}{2}f''(\xi)\cdot\frac{1}{3}(x-c)^2\Big|_a^b \text{ for some } \xi\in(a,b) \\ &= I_0(f)+\frac{(b-a)^3}{24}f''(\xi) \text{ for some } \xi\in(a,b)\end{aligned}$$

  > **积分中值定理:**
  > 若 f 在 $[a,b]$ 上连续，g 在 $[a,b]$ 上不变号，则我们有:
  >
  > $$\int_a^b f(x)g(x)\mathrm{d}x = f(\xi)\int_a^b g(x)\mathrm{d}x \text{ for some } \xi\in(a,b)$$

  因此我们有:

  $$E_0(f)=\int_a^b f(x)\mathrm{d}x - I_0(f) = \frac{(b-a)^3}{24}f''(\xi) \text{ for some } \xi\in(a,b)$$

Solution:
记 $m=\frac{a+b}{2}$ 和 $\delta=\frac{1}{2}h=\frac{b-a}{3}$，则我们有:

$$\begin{array}{c}\dfrac{2a+b}{3}=m-\delta \\ \dfrac{a+2b}{3}=m+\delta \\ \hline f(m-\delta)=f(m)-\delta f'(m)+\dfrac{\delta^2}{2}f''(\xi_1) \text{ for some } \xi_1\in(\dfrac{2a+b}{3},m) \\ f(m+\delta)=f(m)+\delta f'(m)+\dfrac{\delta^2}{2}f''(\xi_2) \text{ for some } \xi_1\in(m,\dfrac{a+2b}{3}) \\ \displaystyle\int_a^b f(x)\mathrm{d}x=(b-a)f(m)+\frac{(b-a)^3}{24}f''(\eta) \text{ for some } \eta\in(a,b)\end{array}$$

于是我们有:

$$
\begin{aligned}
|E(f)| &= \left|\int_a^b f(x)\mathrm{d}x - \frac{b-a}{2}[f(m-\delta)+f(m+\delta)]\right| \\
&= \left|\left[(b-a)f(m) + \frac{(b-a)^3}{24}f''(\eta)\right] - \left[(b-a)f(m) + \frac{b-a}{2}\cdot\frac{(b-a)^2}{8}(f''(\xi_1)+f''(\xi_2))\right]\right| \\
&= \left|\frac{(b-a)^3}{144}(6f''(\eta) - f''(\xi_1) - f''(\xi_2))\right| \\
&\le \frac{(b-a)^3}{18}\max_{x\in[a,b]}|f''(x)|
\end{aligned}
$$

因此代数精度为 1

Part (b)

By partitioning $[a,b]$ into n subintervals of equal length
and applying the quadrature rule from part (a) on each subinterval,
we obtain a composite quadrature rule.
Implement such a composite quadrature rule
and determine the asymptotic behavior of the error $(O(h),\ O(h^2),\ldots)$ when integrating

$$
\int_0^\pi \sin(x)\mathrm{d}x = 2
$$

where $h := (b-a)/n$

Solution:
设 $x_0 = a, x_n = b$
记 $x_i = x_0 + ih$
单个区间 $[x_i, x_{i+1}]$ 上的截断误差为:

$$
\begin{aligned}
|E_i| &= \left|\int_{x_i}^{x_{i+1}} f(x)\mathrm{d}x - \frac{h}{2}\left[f\left(\frac{2x_i+x_{i+1}}{3}\right) + f\left(\frac{x_i+2x_{i+1}}{3}\right)\right]\right| \\
&= \frac{h^3}{18}|f''(\xi_i)| \text{ for some } \xi_i \in [x_i, x_{i+1}] \\
&\le \frac{h^3}{18}\max_{0\le x\le\pi}|f''(x)| \quad (\text{note that } f(x)=\sin(x)) \\
&= \frac{h^3}{18}\max_{0\le x\le\pi}|-\sin(x)| \\
&= \frac{h^3}{18}\cdot 1 \\
&= \frac{h^3}{18}
\end{aligned}
$$

对上式求和可得:

$$
\begin{aligned}
|E| &= \left|\sum_{i=0}^{n-1} E_i\right| \\
&\le \sum_{i=0}^{n-1}|E_i| \\
&\le n\cdot\frac{h^3}{18} \quad (\text{note that } nh = b-a) \\
&= \frac{h^2}{18}(b-a) \\
&= O(h^2)
\end{aligned}
$$

Problem 4

Part (a)

Suppose that $f:[a,b]\mapsto\mathbb{R}$ is twice continuously differentiable.
Show that the remainders (i.e., truncation errors) of the composite midpoint rule and the composite rule are given by

$$K(b-a)h^2|f''(\xi)|$$

for some $\xi\in(a,b)$, where $K=1/24$ for the midpoint rule,
and $K=1/12$ for the trapezoidal rule.

- **Lemma: 中点公式 $I_0(f):=(b-a)f\left(\frac{b-a}{2}\right)$ 的截断误差分析:**
  设 f 在 $[a,b]$ 上二阶连续可导.
  在中点 $c=\frac{a+b}{2}$ 处对 f 进行二阶 Taylor 展开:

  $$f(x)=f(c)+f'(c)(x-c)+\frac{f''(\eta_x)}{2}(x-c)^2 \text{ for some } \eta_x\in(a,b)$$

  左右同时在 $[a,b]$ 上积分，可得:

  $$\begin{aligned}\int_a^b f(x)\mathrm{d}x&=\int_a^b\left\{f(c)+f'(c)(x-c)+\frac{f''(\eta_x)}{2}(x-c)^2\right\}\mathrm{d}x\\&=f(c)(b-a)+f'(c)\cdot\frac{1}{2}(x-c)^2\Big|_a^b+\frac{1}{2}\int_a^b f''(\eta_x)(x-c)^2\mathrm{d}x\quad\text{(use Mean Value Theorem for Integrals)}\\&=I_0(f)+f'(c)\cdot 0+\frac{1}{2}f''(\xi)\int_a^b(x-c)^2\mathrm{d}x\text{ for some }\xi\in(a,b)\\&=I_0(f)+0+\frac{1}{2}f''(\xi)\cdot\frac{1}{3}(x-c)^2\Big|_a^b\text{ for some }\xi\in(a,b)\\&=I_0(f)+\frac{(b-a)^3}{24}f''(\xi)\text{ for some }\xi\in(a,b)\end{aligned}$$

  > **积分中值定理:**
  > 若 f 在 $[a,b]$ 上连续，g 在 $[a,b]$ 上不变号，则我们有:
  >
  > $$\int_a^b f(x)g(x)\mathrm{d}x=f(\xi)\int_a^b g(x)\mathrm{d}x\text{ for some }\xi\in(a,b)$$

  因此我们有:

  $$E_0(f)=\int_a^b f(x)\mathrm{d}x-I_0(f)=\frac{(b-a)^3}{24}f''(\xi)\text{ for some }\xi\in(a,b)$$

- **Lemma: 梯形公式 $I_1(f):=\frac{b-a}{2}[f(a)+f(b)]$ 的截断误差分析: (存疑)**
  设 f 在 $[a,b]$ 上二阶连续可导.
  在区间端点 a 处对 f 进行二阶 Taylor 展开:

  $$f(x)=f(a)+f'(a)(x-a)+\frac{f''(\eta_x)}{2}(x-a)^2\text{ for some }\eta_x\in(a,b)$$

  左右同时在 $[a,b]$ 上积分，可得:

$$
\begin{aligned}
\int_a^b f(x)\mathrm{d}x &= \int_a^b \left\{ f(a) + f'(a)(x-a) + \frac{f''(\eta_x)}{2}(x-a)^2 \right\}\mathrm{d}x \\
&= f(a)(b-a) + f'(a)\cdot\frac{1}{2}(x-a)^2\Big|_a^b + \frac{1}{2}\int_a^b f''(\eta_x)(x-a)^2\mathrm{d}x \text{ (use Mean Value Theorem for Integrals)} \\
&= f(a)(b-a) + f'(a)\cdot\frac{1}{2}(b-a)^2 + \frac{1}{2}f''(\xi_a)\int_a^b (x-a)^2\mathrm{d}x \text{ for some } \xi_a \in (a,b) \\
&= f(a)(b-a) + f'(a)\cdot\frac{1}{2}(b-a)^2 + \frac{1}{2}f''(\xi_a)\cdot\frac{1}{3}(x-a)^2\Big|_a^b \text{ for some } \xi_a \in (a,b) \\
&= f(a)(b-a) + f'(a)\cdot\frac{1}{2}(b-a)^2 + \frac{(b-a)^3}{6}f''(\xi_a) \text{ for some } \xi_a \in (a,b)
\end{aligned}
$$

> **积分中值定理:**
> 若 f 在 $[a,b]$ 上连续，g 在 $[a,b]$ 上不变号，则我们有:
>
> $$\int_a^b f(x)g(x)\mathrm{d}x = f(\xi)\int_a^b g(x)\mathrm{d}x \text{ for some } \xi \in (a,b)$$

交换 a, b 并取负可得:

$$\int_a^b f(x)\mathrm{d}x = f(b)(b-a) - f'(b)\cdot\frac{1}{2}(b-a)^2 + \frac{(b-a)^3}{6}f''(\xi_b) \text{ for some } \xi_b \in (a,b)$$

于是我们有: **(存疑)**

$$
\begin{aligned}
\int_a^b f(x)\mathrm{d}x &= \frac{1}{2}\left[f(a)(b-a) + f'(a)\cdot\frac{1}{2}(b-a)^2 + \frac{(b-a)^3}{6}f''(\xi_a) \right] \\
&\quad + \frac{1}{2}\left[f(b)(b-a) - f'(b)\cdot\frac{1}{2}(b-a)^2 + \frac{(b-a)^3}{6}f''(\xi_b) \right] \\
&= \frac{b-a}{2}(f(a)+f(b)) + \frac{(b-a)^2}{4}(f'(a)-f'(b)) + \frac{(b-a)^3}{12}(f''(\xi_a)+f''(\xi_b)) \\
&= \frac{b-a}{2}(f(a)+f(b)) - \frac{(b-a)^3}{4}f''(\eta) + \frac{(b-a)^3}{12}(f''(\xi_a)+f''(\xi_b)) \text{ for some } \eta \in (a,b) \\
&= I_1(f) - \frac{(b-a)^3}{12}f''(\xi) \text{ for some } \xi \in (a,b)
\end{aligned}
$$

因此我们有:

$$
\begin{aligned}
E_1(f) &= \int_a^b f(x)\mathrm{d}x - I_1(f) \\
&= -\frac{(b-a)^3}{12}f''(\xi) \text{ for some } \xi \in (a,b) \\
&= -\frac{h^3}{12}f''(\xi) \text{ for some } \xi \in (a,b) \\
&\text{where } h = (b-a)
\end{aligned}
$$

Solution:
设 $x_0 = a, x_n = b$
记 $x_i = x_0 + ih$
复化中点公式:

$$M_n = h\sum_{i=0}^{n-1} f(x_{i+\frac{1}{2}}) = h\sum_{i=0}^{n-1} f\left(\frac{x_i + x_{i+1}}{2}\right)$$

在每段小区间 $[x_i, x_{i+1}]$ 上我们都有:

$$\int_{x_i}^{x_{i+1}} f(x)\mathrm{d}x = hf(x_{i+\frac{1}{2}}) + \frac{h^3}{24}f''(\eta_i) \quad x_i < \eta_i < x_{i+1}$$

对上式求和可得:

$$
\begin{aligned}
I(f) &= \int_a^b f(x)\mathrm{d}x \\
&= \sum_{i=0}^{n-1}\int_{x_i}^{x_{i+1}} f(x)\mathrm{d}x \\
&= M_n + \frac{h^3}{24}\sum_{i=0}^{n-1} f''(\eta_i) \\
&= M_n + \frac{h^3}{24}\cdot n\cdot f''(\eta) \text{ for some } \eta\in(a,b) \\
&= M_n + \frac{(b-a)h^2}{24} f''(\eta) \text{ for some } \eta\in(a,b)
\end{aligned}
$$

于是我们有:

$$
|I(f)-M_n| \leq \frac{(b-a)h^2}{24}\max_{x\in[a,b]}|f''(x)|
$$

复化梯形公式:

$$
T_n = \frac{h}{2}\left[f(a)+f(b)+2\sum_{i=1}^{n-1} f(a+ih)\right]
$$

在每段小区间 $[x_i, x_{i+1}]$ 上我们都有:

$$
\int_{x_i}^{x_{i+1}} f(x)\mathrm{d}x = \frac{h}{2}[f(x_i)+f(x_{i+1})] - \frac{h^3}{12}f''(\eta_i) \quad x_i<\eta_i<x_{i+1}
$$

对上式求和可得:

$$
\begin{aligned}
I(f) &= \int_a^b f(x)\mathrm{d}x \\
&= \sum_{i=0}^{n-1}\int_{x_i}^{x_{i+1}} f(x)\mathrm{d}x \\
&= T_n - \frac{h^3}{12}\sum_{i=0}^{n-1} f''(\eta_i) \\
&= T_n - \frac{h^3}{12}\cdot n\cdot f''(\eta) \text{ for some } \eta\in(a,b) \\
&= -\frac{(b-a)h^2}{12} f''(\eta) \text{ for some } \eta\in(a,b)
\end{aligned}
$$

于是我们有:

$$
|I(f)-T_n| \leq \frac{(b-a)h^2}{12}\max_{x\in[a,b]}|f''(x)|
$$

Part (b)

Suppose that $f:[a,b]\mapsto\mathbb{R}$ is twice continuously differentiable.
Suppose that we need to use the composite midpoint rule to approximate $\int_0^1 e^x\mathrm{d}x$.
Determine how many sampling points do we need in order to obtain six correct digits after the decimal point based on a priori error estimate.

Solution:
注意到:

$$\max_{x\in[0,1]} |f''(x)| = \max_{x\in[0,1]} |e^x| = e$$

令误差上界 $\frac{h^2}{24}\max_{x\in[0,1]}|f''(x)| = \frac{e}{24n^2} \leq 0.5\times 10^{-6}$ 解得:

$$n \geq \sqrt{\frac{e}{12\times 10^{-6}}} \approx 476$$

因此需要将 $[0,1]$ 划分为至少 476 个等距区间.

Problem 5

Use Romberg's method to compute $\int_0^1 e^x \mathrm{d}x$ and $\int_0^1 x^{3/2}\mathrm{d}x$.
Keep a record for intermediate results.
What can you say about the convergence?

Solution:

```
% 定义被积函数和积分区间
f1 = @(x) exp(x);
a1 = 0; b1 = 1;
f2 = @(x) x.^(3/2);
a2 = 0; b2 = 1;

% 设置最大二分次数（k_max=3 包含k=0,1,2,3）
k_max = 3;

% 计算Romberg表
R1 = romberg_integration(f1, a1, b1, k_max);
R2 = romberg_integration(f2, a2, b2, k_max);

% 显示积分一的中间结果
disp('积分一: \int_0^1 e^x dx 的Romberg表: ');
print_romberg_table(R1, k_max);
disp('精确值: e - 1 ≈ 1.718281828459045');

% 显示积分二的中间结果
disp('积分二: ∫₀¹ x^(3/2) dx 的Romberg表: ');
print_romberg_table(R2, k_max);
disp('精确值: 0.4');

function R = romberg_integration(f, a, b, k_max)
    R = zeros(k_max+1, k_max+1);
    % 计算第一列（m=0）
    for k = 0:k_max
        n = 2^k;
        h = (b - a)/n;
        x = a + h*(0:n);
        fx = f(x);
        R(k+1, 1) = h * (0.5*fx(1) + sum(fx(2:end-1)) + 0.5*fx(end));
    end
    % Richardson外推
    for m = 1:k_max
        for k = m:k_max
            R(k+1, m+1) = R(k+1, m) + (R(k+1, m) - R(k, m)) / (4^m - 1);
        end
    end
end
```

```
function print_romberg_table(R, k_max)
    % 打印表头
    header = 'k/m | ';
    for m = 0:k_max
        header = [header, sprintf('m=%-9d ', m)];
    end
    disp(header);
    disp(repmat('-', 1, 12*(k_max+2)));

    % 打印每行数据
    for k = 0:k_max
        row_str = sprintf('%2d | ', k);
        for m = 0:k_max
            if m <= k
                row_str = [row_str, sprintf('%-10.8f ', R(k+1, m+1))];
            else
                row_str = [row_str, sprintf('%-10s ', ' ')];
            end
        end
        disp(row_str);
    end
end
```

运行结果:

```
积分一：∫₀¹ e^x dx 的Romberg表：
k/m | m=0         m=1         m=2         m=3
------------------------------------------------------------
 0 | 1.85914091
 1 | 1.75393109 1.71886115
 2 | 1.72722190 1.71831884 1.71828269
 3 | 1.72051859 1.71828415 1.71828184 1.71828183
精确值：e - 1 ≈ 1.718281828459045
积分二：∫₀¹ x^(3/2) dx 的Romberg表：
k/m | m=0         m=1         m=2         m=3
------------------------------------------------------------
 0 | 0.50000000
 1 | 0.42677670 0.40236893
 2 | 0.40701811 0.40043192 0.40030278
 3 | 0.40181246 0.40007725 0.40005361 0.40004965
精确值：0.4
```

收敛性分析:

- ① e^x 是光滑函数，Romberg 外推迅速收敛至精确值
- ② $x^{3/2}$ 在 $x=0$ 处二阶导数发散，导致收敛速度显著减慢

Problem 6 (optional)

Linear and quadratic Bézier curves can exactly fit line segments and parabolas, respectively.
However, many computer programs only support cubic Bézier curves.
Suppose you want to plot a segment of parabola of the form $y = ax^2 + bx + c$ for $x \in [x_1, x_2]$.
How to construct four control points for this purpose?